

Team ROC Project Workbook

Trajectory Integrated Decision Engine for Heart-Failure (TIDE-HF) — Phase 1A

Members

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Client

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1. Meeting Log

Weekly check-ins with Dr. Ionita ran from late February through mid-April. Notes for each meeting live in the Meetings/ folder; the summary below is the project-relevant gist of each.

- 2/26 — Introductory meeting. Established project scope and clinical goals.
- 3/5 — Weekly check-in. Scope clarification: focus on chronic HFrEF, NYHA C/D, four GDMT classes.
- 3/13 — Weekly check-in. Dataset discussion: agreed to pursue MIMIC-IV + MIMIC-IV-ECG.
- Spring break.
- 3/26 — Weekly check-in. Initial dataset findings and titration-method discussion.
- 4/2 — Weekly check-in. First random-forest classifier results and limitations.
- 4/14 — Weekly check-in. Synthetic-data plan, engine v1 sketched, strategy modes outlined.

2. Project Description

Project Title: Trajectory Integrated Decision Engine for Heart-Failure (TIDE-HF) — Phase 1A

Patient-facing app providing post-discharge (from a heart failure event) decision support utilizing patient data from wearables, Epic hospital record data, and hardcoded “guardrails” (predefined limits from existing case studies). The goal of this agent is to provide home-monitoring and execution of bounded clinical actions: preemptive emergency alerts, medication dosage alterations, and lab orders.

Phase 1A scope

Full autonomy for non-medication workflows. Decisions on medication changes are treated as proposals to be reviewed by clinicians.

- HF classification: Chronic HFrEF, NYHA Class C/D (relevant to titration decisions).
- Medication classes to focus on: Beta-blockers, RAAS inhibitors, Diuretics, SGLT-2 inhibitors.
- Sensors to be used (emulated in data): scale, BP/HR monitor, single-lead ECG patch, Zoll inductance patch.
- Metrics: weight, BP, HR, O2 saturation, ECG, induction (water retention).
- Main priority: logic framework for medication titration (strict guideline-following).
- Secondary: data collection, synthetic data creation, ML/pattern-recognition development.
- Loft goal: reinforcement-learning value function to learn the best path for titration.

Initial research plan

After our first meetings with Dr. Ionita, the team performed preliminary research across four areas: the heart-failure clinical domain, existing decision-support solutions, titration methods and guidelines, and publicly available datasets. Each of those is documented in the sections that follow.

3. Domain Research

Sources: Ninja Nerd YouTube videos on Congestive Heart Failure and Drugs for Heart Failure. These cover the pathophysiology, the consequences of left vs. right HF, and the mechanism of action of each major drug class. Notes below are the team's working summary.

Congestive heart failure pathophysiology

Key modes: Left HF, Right HF, HFrEF (reduced ejection fraction $\leq 35\%$, typically systolic), HFpEF (preserved, typically diastolic). The project focuses on HFrEF for the titration decisions because that is where GDMT evidence is strongest. The HFpEF cohort should still benefit from outcome monitoring.

Key equation: $BP = CO \times SVR$.

Left-heart complications:

- Pulmonary congestion / pulmonary edema. Reduced cardiac output backs blood up into pulmonary arteries, causing fluid to leak into the lungs.
- Symptoms: dyspnea on exertion, at rest, or supine; hypoxia in severe cases.
- Cardiogenic shock. Reduced CO drives up systemic vascular resistance and causes organ malperfusion (brain, coronary, kidneys, GI).
- Symptoms: cold, pale, mottled extremities; encephalopathy; TIA / CVA; myocardial infarction; acute kidney injury; acute mesenteric ischemia.

Right-heart complications:

- Increased central venous pressure / peripheral edema. Excess blood backs up into the jugular vein and lower extremities; hepatic congestion.
- Symptoms: distended jugular vein, lower-extremity edema, ascites, liver failure / cirrhosis in extreme cases.
- Cardiogenic shock from right-ventricle dilation reducing left-ventricle volume and CO.

Heart-failure medications

Reduced BP triggers compensation: norepinephrine reaches β_1/α_1 receptors in the heart, arteries, veins, and kidneys. The kidney response (RAAS) produces renin, then AT-I (combined with ACE), then AT-II, which constricts arteries and veins, raises ADH and aldosterone, retains sodium and water, and ultimately stiffens the kidney filter membrane. Decreased CO also raises BNP (which blocks AT-II) and produces backflow congestion in the lungs (left HF) or the venous system (right HF). Compensation increases HR, contractility, and O₂ demand; it also accelerates LV hypertrophy and dilation. The drug classes below each intervene at one or more points in this cascade.

- Alpha/Beta-blockers (first line). Block A1/B1 receptors. Reduce HR, increase diastole, reduce contractility, inhibit RAAS. Initial drop in EF, eventual increase.
- Good for: HFrEF, post-MI, CAD.
- Adverse effects: bradycardia, decreased EF (transient), bronchospasm, masked hypoglycemia.
- RAAS inhibitors (first line). Block production or absorption of AT-II. Arteries dilate; veins dilate; ADH and aldosterone fall; sodium and water excretion increases.
- Good for: HFrEF, HFpEF, hypertension.
- Adverse effects: increased creatinine, hyperkalemia, bradykinin-driven cough, angioedema.
- ARNi (alternate to ACEi or ARB). Increases BNP and adds RAAS inhibition; significantly reduces AT-II.
- Good for: HFrEF, HFpEF, HTN.
- Adverse effects: increased creatinine, hyperkalemia, lower BP than ACEi/ARB, cough, angioedema. Cannot be co-administered with ACEi (36-hour washout).
- Vasodilators (alternative to ACEi or ARB). Arteries and veins dilate; forward flow up, congestion down.
- Good for: patients with decreased renal function.
- Adverse effects: reflex tachycardia.
- Diuretics (symptom control). Block sodium and water reabsorption; blood volume falls; preload falls; pulmonary and peripheral edema fall.
- Good for: dyspnea, hypoxic respiratory failure, lower-body edema, fluid retention.
- Adverse effects: hyponatremia, hypokalemia, metabolic alkalosis, hyperuricemia.

- SGLT2 inhibitors. Block sodium-glucose co-transport in the kidney; mild diuresis, blood-glucose excretion, cardioprotection independent of glycemic effect.
- Good for: HFrEF, HFpEF, type 2 diabetes with cardiovascular risk.
- Adverse effects: volume depletion, genital mycotic infection, euglycemic DKA, transient eGFR dip on initiation.
- Ivabradine (refractory HFrEF on max beta-blocker). Blocks funny-sodium channels in the SA node; lowers HR; lengthens diastole.
- Good for: refractory HFrEF in normal sinus rhythm.
- Adverse effects: bradycardia, transient brightness / phosphenes.
- Inotropes (Digoxin; dobutamine and milrinone for cardiogenic shock only).
- Digoxin: good for refractory HFrEF with AFib; adverse effects include arrhythmia and digoxin toxicity (worse with hypokalemia).
- Dobutamine / milrinone: only for cardiogenic shock, not outpatient.

4. Existing Solutions and Titration Methods

Three existing platforms are most relevant to this project: AIM-POWER (BiovitalsHF), STRONG-HF, and VITAL-HF. The Greene 2021 “Better, Faster, Stronger” commentary and the Optimal Implementation of GDMT for Heart Failure talk from the 2025 Emory Cardiology Conference were the two clearest summaries we found of the rapid-sequence approach.

AIM-POWER (BiovitalsHF)

Digital platform that generates GDMT titration recommendations for a clinician to review every two weeks. Not patient-facing. Patients are issued an iHealth Lite scale and a Welch Allyn BP/HR monitor; weight is measured daily, BP/HR twice daily; labs are ordered by site clinicians on a regular cadence. The proprietary algorithm uses average weekly BP, average weekly HR, and lab values to either initiate a new medication or recommend up-titration. When multiple recommendations are possible, initiating new medications takes priority over up-titrating existing ones.

Patient selection (verbatim from the trial protocol):

“Patients at least 22 years old who had symptomatic HFrEF (i.e., left ventricular ejection fraction $\leq$ 40% within 12 months) and who were not yet optimized on medical therapy for HFrEF were eligible for enrollment. Potential for further titration of medical therapy was determined by local investigators and could have ranged from participants on no treatments for HFrEF to those on multiple medications but with at least one at $\leq$ 50% of target dose. Participants were enrolled as outpatients and could not have

had an HF hospitalization within the prior 28 days. Key exclusion criteria were absolute contraindications for HFrEF therapy, a baseline creatinine of > 2.0 mg/dL or an estimated GFR < 30 mL/min/1.73 m², a potassium > 5.0 mEq/L, and a baseline systolic blood pressure < 100 mm Hg.”

Outcomes and complications monitored:

- Cardiovascular death
- HF hospitalization
- Change in optimal therapy score
- Change in Kansas City Cardiomyopathy Questionnaire
- Change in NT-proBNP
- Acute kidney injury
- Hypotension
- Hyperkalemia
- Angioedema
- Bradycardia
- Syncope

STRONG-HF

Clinical trial designed to achieve 100% of guideline-recommended dose within two weeks of discharge. Acute-HF hospitalization plus immediate post-discharge care pathway. AHA HF management guidelines (2022) provide initial and target doses for all relevant medications. STRONG-HF showed a clear reduction in 180-day all-cause death or HF readmission compared to usual care, but at the cost of a meaningfully higher visit cadence and a higher rate of minor adverse events (mostly hypotension and renal). The team uses STRONG-HF as the evidence base for the rapid-sequence strategy in the engine.

VITAL-HF (Story Health)

Remote data collection for physicians, following the STRONG-HF titration path. Notable because it is the closest to a live deployment of rapid-sequence titration in the US outpatient setting.

Other references

- Optimal Implementation of GDMT for Heart Failure (2025 Emory Cardiology Conference) — more conservative than STRONG-HF; closer to a traditional ladder.
- Detecting Heart Failure Using Wearables: A Pilot Study — wearable-only HF detection.
- Greene 2021, “Initiation and Up-titration of Guideline-directed Medical Therapy for Patients with Heart Failure: Better, Faster, Stronger!” — plain-language summary of rapid-sequence rationale.

- In-hospital Initiation and Up-titration of GDMT for HFrEF (PMC).
- Patient profiling in heart failure for tailoring medical therapy (HFA-ESC consensus document).
- Drug Layering in Heart Failure: Phenotype-Guided Initiation.

5. Adverse Effects to Monitor

The classifier is supposed to flag plausible adverse-effect onset between scheduled lab draws so the engine can decide whether to hold a class, order labs, or proceed with titration. The table below lists the major adverse effects per GDMT class along with the signals that the classifier and the engine watch for them.

Class	Representative agents	Adverse effects	Watch for
ACEi	Lisinopril, Enalapril, Ramipril	Hyperkalemia, AKI / rising creatinine, hypotension, dry cough, angioedema	K, Cr/eGFR, SBP, lip/tongue swelling history
ARB	Losartan, Valsartan	Hyperkalemia, AKI, hypotension. Lower angioedema risk than ACEi but possible	K, Cr/eGFR, SBP
ARNi	Sacubitril/Valsartan	Hyperkalemia, AKI, hypotension (more symptomatic than ACEi/ARB), angioedema	K, Cr/eGFR, SBP, ACEi-washout (36 h), angioedema history
Beta blocker	Carvedilol, Metoprolol succinate, Bisoprolol	Bradycardia, AV block, hypotension, fatigue, transient HF worsening on initiation, bronchospasm, masked hypoglycemia	HR, PR interval, weight gain, dyspnea, asthma history
MRA	Spironolactone, Eplerenone	Hyperkalemia (risk rises sharply with eGFR < 30), AKI, gynecomastia (spironolactone)	K, Cr/eGFR; avoid if K > 5.0 or eGFR < 30
SGLT2i	Dapagliflozin, Empagliflozin	Volume depletion, hypotension (esp. with	SBP, weight, urinary symptoms, ketones if

		loops), genital mycotic infection, UTI, euglycemic DKA, transient eGFR dip, rare Fournier's gangrene	unwell, eGFR (avoid initiating if < 20)
Loop diuretic	Furosemide, Torsemide, Bumetanide	Hypokalemia, hypomagnesemia, hyponatremia, volume depletion, prerenal AKI, ototoxicity, hyperuricemia / gout, metabolic alkalosis	K, Mg, Na, weight loss rate, Cr
Thiazide diuretic	HCTZ, Chlorthalidone, Metolazone	Hyponatremia (more than loops), hypokalemia, hypercalcemia, hyperuricemia, hyperglycemia, dyslipidemia, photosensitivity rash	Na (especially in elderly), K, glucose

Cross-cutting notes for the titration engine:

- Hyperkalemia and renal dysfunction gate all four RAAS-axis classes (ACEi, ARB, ARNi, MRA). The engine holds them together when K rises or Cr jumps more than 30% above baseline.
- Hypotension gates ACEi/ARB/ARNi, beta-blocker, and SGLT2i. It is exacerbated by aggressive loop dosing.
- Volume depletion is the shared signal that tells the engine to back off SGLT2i and the loop simultaneously.
- Bradycardia is essentially specific to beta-blocker (and to ivabradine, which the current engine does not cover).
- Severe hypotension (SBP < 80), confirmed K > 6.0, and any emergency flag trigger a global stop with all GDMT held and an escalation prompt.

These map onto the 11 classifier flag columns trained:

Flag	What it captures
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hypotension_detected	SBP trending below 100 mmHg
severe_hypotension_detected	SBP below 80 — global stop
bradycardia_detected	HR trending below 50 bpm
severe_bradycardia_detected	HR below 40 — global stop
volume_depletion_detected	Weight loss + hypotension + rising HR
worsening_HF_detected	Weight gain + SpO2 drop + dyspnea pattern
hyperkalemia_detected	ECG signs (peaked T, wide QRS) without lab confirmation
renal_dysfunction_detected	Cr or eGFR drift suggesting AKI
hyponatremia_detected	Patterns suggestive of low sodium
metabolic_acidosis_detected	Patterns suggestive of low bicarbonate
any_emergency_flag	Catch-all severe combination — global stop

6. Dataset Selection and Cleaning

The team needed two categories of patient data:

- Patient signal data: ECG waveforms, blood pressure, heart rate, oxygen saturation (SpO2), and impedance-patch data for fluid retention.
- Medication data: beta-blockers, RAAS inhibitors (ACEi / ARB / ARNi), diuretics, SGLT-2 inhibitors.

Datasets considered

- MUSIC dataset (PhysioNet): 936 CHF patients, 24-hour Holter ECGs, medications, LVEF.
- BIDMC CHF Database (PhysioNet): 15 patients, ICU waveforms (ECG, BP, SpO2).
- Zigong dataset (PhysioNet): 2,008 HF patients, BP, HR, medications, post-discharge outcomes at 28 d / 3 mo / 6 mo.
- MIMIC-IV: electronic health records from Beth Israel Deaconess Medical Center (2008–2019), 546,028 patient admissions — vitals, labs, medications, diagnoses, clinical notes.
- MIMIC-IV-ECG: ~800,000 12-lead 10-second 500 Hz ECG recordings from ~160,000 patients.

We finalized MIMIC-IV (which meets most of the project requirements but lacks waveform ECG) combined with MIMIC-IV-ECG to provide the ECG component. Required PhysioNet credentialing was completed by all team members.

ICD code selection

Two code sets were curated. The first (filters_v1) is a strict CHF cohort built from the systolic / combined-systolic-and-diastolic codes, used for the initial classification experiments. The second (filters_v2) expands the cohort and adds adverse-effect and contraindication codes so that the synthetic-data pipeline can sample realistic patient contexts.

Strict CHF cohort (filters_v1, 19 codes):

ICD code	Version	Long title
I50	10	Heart failure
I502	10	Systolic (congestive) heart failure
I5020–I5023	10	Unspecified / acute / chronic / acute-on-chronic systolic HF
I504	10	Combined systolic and diastolic heart failure
I5040–I5043	10	Unspecified / acute / chronic / acute-on-chronic combined HF
42820–42823	9	Systolic heart failure (ICD-9)
42840–42843	9	Combined systolic and diastolic heart failure (ICD-9)

Expanded cohort with adverse effects and contraindications (filters_v2):

Category	Codes	Notes
CHF (broader)	I501–I509, I508, I509, 4280, 4281, 42830–42843	Includes diastolic, left-failure-only, and unspecified HF in addition to filters_v1.
Hyperkalemia	E875, 2767	Hyperkalemia / hyperpotassemia.

Renal insufficiency	N17, N18, N19, 584, 585, 586	AKI, CKD, unspecified renal failure.
Hypotension	I95–I952, 4580, 4589	Idiopathic, orthostatic, drug-induced, unspecified.
Bradycardia	R001, 42789	Bradycardia and other dysrhythmias.
Ketoacidosis	E101, E111, E131, 24910, 25010	Diabetic ketoacidosis, all types.
Edema	R60, J81, 7823	Generalized and pulmonary edema.
Hyponatremia	E871, 2761	Hyposmolality and hyponatremia.
Hypernatremia	E870, 2760	Hypernatremia.
Acidosis	E872, 2762	Metabolic acidosis.
Pulmonary edema	J810	Acute pulmonary edema.
Dyspnea	R0600–R0602	Dyspnea / orthopnea / shortness of breath.
Contraindications	T783, O00, O9A, Z33–Z39, V22–V24, I701, K70–K76, J45, J46, J9801, I440–I442, I495, R570, N184–N186	Angioedema, pregnancy, renal-artery stenosis, liver disease, asthma, AV block, cardiogenic shock, severe CKD.

Lab itemid selection

Roughly 70 labevents itemids were tagged into the categories listed in `Filters_v1/Labs_codes.csv`. The runtime engine uses a smaller subset — the canonical blood values per analyte — because the others are body-fluid or urine variants that rarely appear in routine HF follow-up.

Analyte	MIMIC itemids	Use in engine
Potassium	50971 (Chemistry, Blood), 50822 (Whole Blood)	Hyperkalemia gate

Creatinine + eGFR	50912 (Cr Chemistry), 50920 (eGFR MDRD), 53161/53180 (eGFR CKD-EPI)	Renal gate
BUN	51006 (Chemistry)	Renal gate, secondary
Sodium	50983 (Chemistry)	Hyponatremia gate (loop AE)
Bicarbonate / pH	50882, 50803, 50820, 50818, 50821	Metabolic acidosis
Glucose	50931, 50809	Ketoacidosis screen
Magnesium / Phosphate	50960, 50970	Loop AE
NTproBNP	50963	HF severity tracking (not used in titration directly)

Files used and class distribution

The first cleaned dataset, master_CHF_visits_v7.parquet (68 MB), was assembled by joining diagnoses_icd, admissions, prescriptions, labevents, and chartevents on hadm_id, filtering to the strict CHF cohort, and per-visit aggregating the relevant vitals and labs. ICD codes for adverse effects were used to label each visit.

Class distribution per binary task on master_CHF_visits_v7 (after balanced sampling):

Adverse effect	Healthy n	Adverse n
Chronic Kidney Disease	2,493	4,100
Hyperkalemia	1,132	1,132
Hypotension	1,760	1,760
Edema	109	109
Bradycardia	155	155
Ketoacidosis	168	168

7. EDA and First Classifier

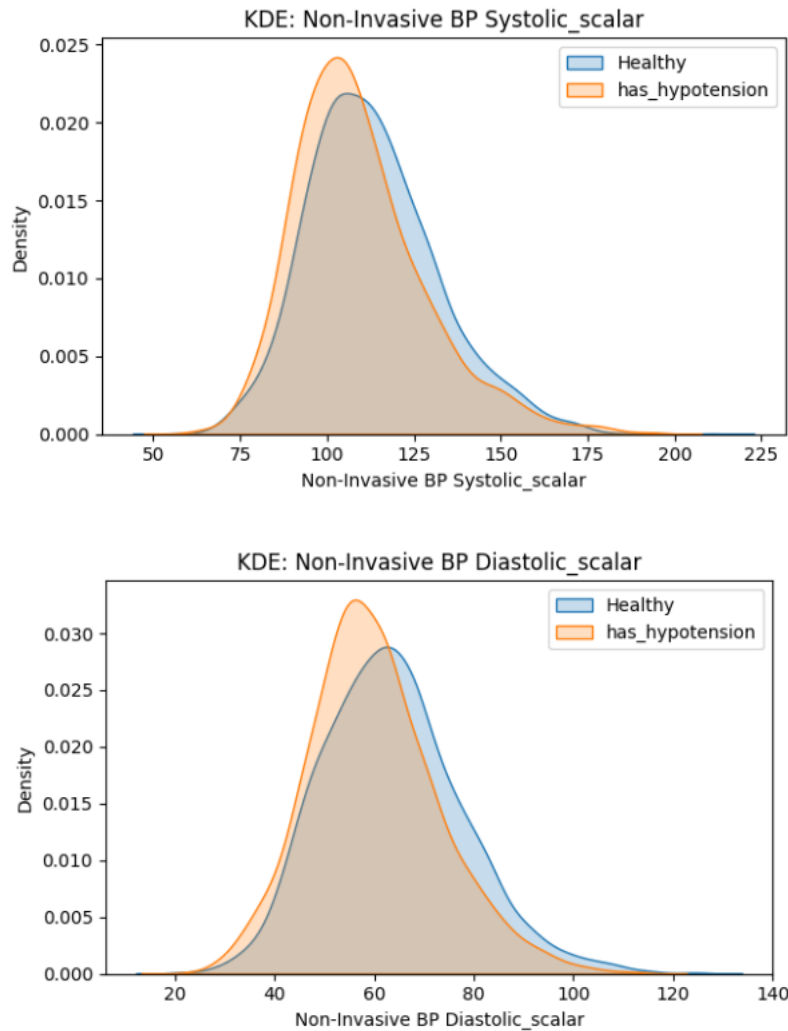
The first classifier was a random forest trained per binary task on visit-level features extracted from the strict CHF cohort. Three feature-extraction strategies were compared: the earliest measurement of each

vital, the median across the visit, and a fuller summary-statistics representation (mean, min, max, range, slope) across the entire visit. All three strategies produced similar results.

Random-forest results (per-class precision/recall/F1):

Adverse effect	Class	Prec.	Rec.	F1	Support	Acc.	Total support
Chronic Kidney Disease	Healthy	0.61	0.16	0.26	624	0.64	1,649
Chronic Kidney Disease	CKD	0.65	0.94	0.77	1,025	0.64	1,649
Hyperkalemia	Healthy	0.56	0.59	0.57	283	0.56	566
Hyperkalemia	Hyperkalemia	0.56	0.53	0.54	283	0.56	566
Hypotension	Healthy	0.57	0.50	0.53	440	0.56	880
Hypotension	Hypotension	0.55	0.62	0.58	440	0.56	880
Edema	Healthy	0.45	0.50	0.47	28	0.44	55
Edema	Edema	0.42	0.37	0.39	27	0.44	55
Bradycardia	Healthy	0.75	0.69	0.72	39	0.73	78
Bradycardia	Bradycardia	0.71	0.77	0.74	39	0.73	78
Ketoacidosis	Healthy	0.69	0.79	0.73	42	0.71	84
Ketoacidosis	Ketoacidosis	0.75	0.64	0.69	42	0.71	84

Some success in detecting ketoacidosis and bradycardia. Detection of all other classes was nearly a coin flip. The diagnosis of the failure came from the EDA on the features themselves: the “healthy” and adverse-effect classes overlap heavily on every vital we tracked. The most glaring example is hypotension — patients diagnosed with hypotension routinely have BP readings well above the 90/60 mmHg textbook bound, because the diagnoses come from emergency-room and ICU visits in which the patient is actively being treated for an acute event. The “healthy” cohort is also a snapshot from an admission rather than a baseline, so the same vital can land in either bucket.



The takeaway: MIMIC-IV gives us excellent demographic, lab, and prescription information, but the vital trajectories embedded in the dataset are not representative of the home-monitoring scenario the project actually targets. The visit-level features were a dead end. We needed synthetic vital trajectories anchored in real MIMIC patient contexts.

8. Synthetic Dataset Creation

The v2 pipeline generates synthetic patient-weeks. Each row is one week of twice-daily vitals and ECG measurements (14 timepoints) for a single patient context. The per-week labels are sampled in advance to enforce balanced incidence across the eleven adverse-effect flags, and the trajectories are then synthesized to be consistent with the sampled labels. The patient context — age, gender, baseline labs, prescription pattern, contraindication history — comes from real MIMIC visits. The result is a training set that preserves realistic clinical correlations while still giving the classifier enough positive examples of every adverse effect to learn from.

Three source artefacts

synthetic_CHF_visits_v2.parquet (135 MB, ~137k visits):

Pre-built snapshot of MIMIC-IV CHF admissions extracted with filters_v2. Each row is one admission, with: subject_id, hadm_id, anchor_age, gender, baseline_labs (a dict of K / Na / Cr / eGFR / HCO3 medians from labs_2d), gdmr_prescriptions (list of records with drug name, dose, target, ratio), labs_2d (the lab events from the 48 hours around the index event), contraindication_diagnoses (list of ICD codes from icd_contraind), and ecg_study_ids (links into MIMIC-IV-ECG when present). This file is the seed for the mimic-mode generator.

synthetic_data_weeks.parquet (the training set):

Produced by the data_gen module. This data is generated using Gaussian sampling for each vital based on the vitals patterns associated with each condition. For each requested patient-week:

- Sample a label vector from a balanced distribution over the 11 flags.
- Either (mimic mode) sample a real visit and copy its demographics, baseline labs, and GDMT prescription state; or (distribution mode) draw all of the above from Gaussians. Mimic mode is the default when synthetic_CHF_visits_v2.parquet is present.
- Synthesize 14 twice-daily timepoints (HR, SBP, DBP, SpO2, weight, plus ECG QRS / QTc / rhythm / T-peaked flag) consistent with the sampled labels. Each label has a deterministic vital signature — e.g. hyperkalemia drives a peaked T wave, widening QRS, and a rhythm shift; volume depletion drives weight loss, falling SBP, and rising HR.
- Optionally synthesize a confirmation lab draw at the end of the week if the label vector implies labs were ordered. The confirmation lab schema mirrors baseline_labs.
- Emit one parquet row per week.

10,000 weeks is the default training-set size. Generation takes about 90 seconds on a modern laptop in mimic mode. The output is then featurized (108 features per week) and fed into the classifier trainer.

chf_weekly_synthetic.csv (GAN generated training set):

Produced by GAN model. Will replace Gaussian-based synthetic dataset for synthetic classification experiments. More realistic data compared to Gaussian sampling, to better train classifier. Vitals, labs, ECG features, GDMT medications, and all 11 adverse effect flags were then generated to be clinically consistent with that scenario: hyperkalemia patients had K⁺ above 5.6, bradycardia patients had HR below 55, worsening HF patients gained 2–8 kg over the week with SpO2 dropping below 93%.

Each patient received 14 vital readings (twice daily × 7 days), with weekly summary statistics — mean, min, max, and trend — computed as the classifier input features. Labs followed the same logic: renal dysfunction patients had creatinine 1.9–4.5 mg/dL with eGFR below 40, and eGFR was calculated from the CKD-EPI formula rather than hardcoded. GDMT prescriptions were assigned at real HFREF rates —

ARNi 45%, beta-blocker 88%, MRA 70%, SGLT2i 65%. After generation, three post-processing fixes were applied: correcting medication prescription rates that were near zero in early versions, ensuring `renal_dysfunction_detected` fired whenever creatinine rise exceeded 30%, and ensuring `bradycardia_detected` fired whenever `HR_min` dropped below 50 BPM. The final output was 137,712 rows with 11 binary AE flag columns as classifier labels and approximately 40 vital, ECG, and medication columns as classifier features.

Resulting label balance

Roughly 30–40% positive rate per flag. This is high relative to the real clinical incidence — deliberately so, because real-world incidence is low enough that a naively trained classifier would learn to always predict “not detected.” The production threshold is then calibrated against the lower real incidence at deployment time. For now the team operates at the default 0.5 threshold.

9. Classifier v2: LightGBM Multi-Label

The second classifier is a set of eleven LightGBM gradient-boosted classifiers, one per adverse-effect flag, trained on the synthetic patient-weeks. The models share a single feature representation but produce independent probability outputs. The engine consumes the binarized flags at threshold 0.5 by default; the strategy applier can also consume the raw probabilities for tie-breaking.

Featurization

Each row’s 14 timepoints are reduced to 108 scalar features. Per vital and per ECG-scalar channel we compute mean, min, max, range, first value, last value, slope, and number of missing samples. Rhythm is one-hot encoded across the 11 categorical rhythm labels. The patient’s age, gender (one-hot M), and per-class on/off / actual-dose / target-dose / dose-ratio fields round out the feature vector.

Training

- GroupShuffleSplit on `subject_id` (no patient appears in more than one of train / val / test).
- 70% train, 15% validation, 15% test.
- LightGBM with `n_estimators=500`, `learning_rate=0.05`, `num_leaves=63`, `min_child_samples=30`, `subsample=0.85`, `colsample_bytree=0.85`, `reg_lambda=1.0`, `early_stopping_rounds=30` on validation AUC.
- Trained eleven independent boosters — one per label — because the labels are not mutually exclusive and the per-label class balance varies.

Per-flag test-set AUROC (10,000 weeks, mimic-mode synth)

Flag	AUROC
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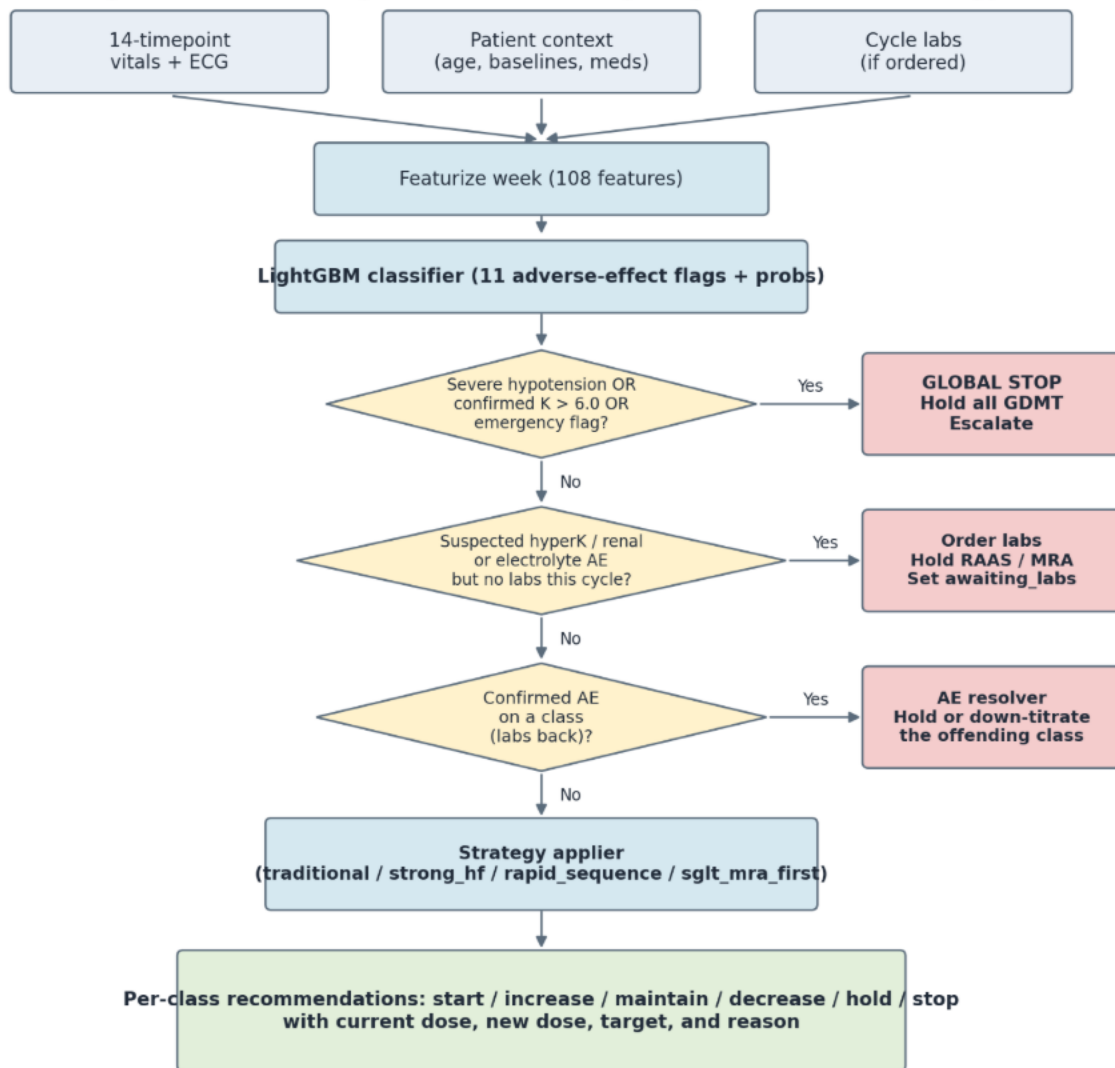
hypotension_detected	0.93
severe_hypotension_detected	0.97
bradycardia_detected	0.95
severe_bradycardia_detected	0.98
volume_depletion_detected	0.91
worsening_HF_detected	0.89
hyperkalemia_detected	0.84
renal_dysfunction_detected	0.81
hyponatremia_detected	0.78
metabolic_acidosis_detected	0.79
any_emergency_flag	0.96

The AUROCs above are representative of a 10,000-week training run; each retraining run reproduces them within ± 0.01 – 0.02 . The two weakest flags — hyponatremia and metabolic acidosis — are the ones with the weakest vital signature in the synthesizer; both are typically confirmed by labs anyway, so a borderline classifier score still results in the right downstream behavior (engine orders labs).

10. Titration Logic

The titration engine sits between the classifier and the strategy applier. Its job is to translate the eleven flag probabilities (plus the patient context, plus any cycle labs) into a per-class engine action: maintain, hold, immediate AE, lab AE, or global stop. Engine v1.2 strictly gates the lab-dependent paths on whether labs are actually available this cycle, and carries an awaiting_labs set across cycles.

Figure 1: Titration logic — classifier, lab gating, AE resolver, strategy



Decision points

- Global stop. Triggered by any of: severe_hypotension_detected, severe_bradycardia_detected, any_emergency_flag, or a confirmed K > 6.0 in the cycle labs. All GDMT held, escalation prompt issued.
- Lab-AE candidate. The classifier has flagged hyperkalemia, renal dysfunction, hyponatremia, or metabolic acidosis, but no labs have been drawn this cycle. The engine holds the gated classes (RAAS-axis for hyperK / renal, loop for sodium issues), records them in awaiting_labs, and asks for labs. The next cycle either confirms the AE or clears the hold.

- Confirmed AE. Labs are back this cycle and confirm an AE on a class that is currently in awaiting_labs. The AE resolver fires:
 - hyperkalemia: hold MRA first, then ARNi, then ACEi/ARB, in that order.
 - renal dysfunction: hold MRA, ACEi/ARB/ARNi; consider loop dose increase if volume overload coexists.
 - volume depletion: stop SGLT2i and decrease loop.
 - bradycardia: down-titrate beta-blocker first.
 - worsening HF: increase loop, hold beta-blocker if symptomatic.
- No AE active. The engine returns maintain on every class and hands off to the strategy applier, which decides whether to start a new class or up-titrate an existing one.

Contraindications

Contraindications are derived once at session start from the patient's ICD history (icd_contraind.csv), baseline labs, and current medication list (meds_contraind.csv). Examples: angioedema history blocks ACEi and ARNi (preferred RAAS falls to ARB); severe asthma blocks beta-blocker; eGFR < 30 derives an MRA contraindication; pregnancy blocks all RAAS agents and SGLT2i; any AV block higher than first degree blocks the beta-blocker.

11. Strategy Selection

Once the engine has decided what each class can safely do this cycle, the strategy applier decides what each class will actually do. We expose four strategies, each matching a published or proposed approach to GDMT initiation, and let the clinician pick.

Strategy	Behavior
traditional	One new class started per week, in the order RAAS → beta-blocker → MRA → SGLT2i → loop. Single-rung up-titration. Closest to the conservative approach described at the 2025 Emory cardiology conference.
strong_hf	All eligible classes started simultaneously, double-rung up-titration. Targets the 100%-of-target-dose-in-two-weeks goal from the STRONG-HF trial.
rapid_sequence	All eligible classes started simultaneously, single-rung up-titration. Greene 2021 compromise:

sglt_mra_first	faster than traditional, less aggressive than STRONG-HF. Phase 1 starts SGLT2i + MRA together; phase 2 (the following cycle, conditional on stable labs) adds ARNi + beta-blocker. Built around the observation that SGLT2i and MRA have the cleanest tolerability profile, so they make the safest first move.
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Each strategy returns a per-class change record with the engine action, the concrete action (start_medication / increase_dose / maintain_dose / decrease_dose / stop_medication / hold_titration), the current dose, the new dose, the dose ratio relative to target, and a free-text reason. The new dose is chosen by stepping along a class-specific dose ladder built at import time from GDMT_drugs_ref.csv: doubling from the initial dose to the target, with single-rung or double-rung steps depending on the strategy.

12. Sample UI

A Streamlit app provides a local demo. It runs entirely on the laptop, with no cloud tunnel, on <http://localhost:8501>.

Controls

- Patient-preset selector with 17 pre-built scenarios that exercise the main decision paths (stable / partially titrated / fully titrated, suspected vs. confirmed adverse effects, and the major contraindications).
- “Resynthesize week” button to redraw the 14 timepoints from the same preset.
- Input mode toggle: summary statistics (HR mean / min, SBP mean / min, SpO2 min, QRS max, QTc max, T-peaked, dominant rhythm, weight delta) or a 14-row CSV of the individual timepoints. Either view can be edited and synced back to the timepoint list.
- Strategy dropdown bound to the four strategies described above.
- Labs-arrived checkbox — controls whether the engine treats this cycle as having labs.
- Compute button.

Outputs

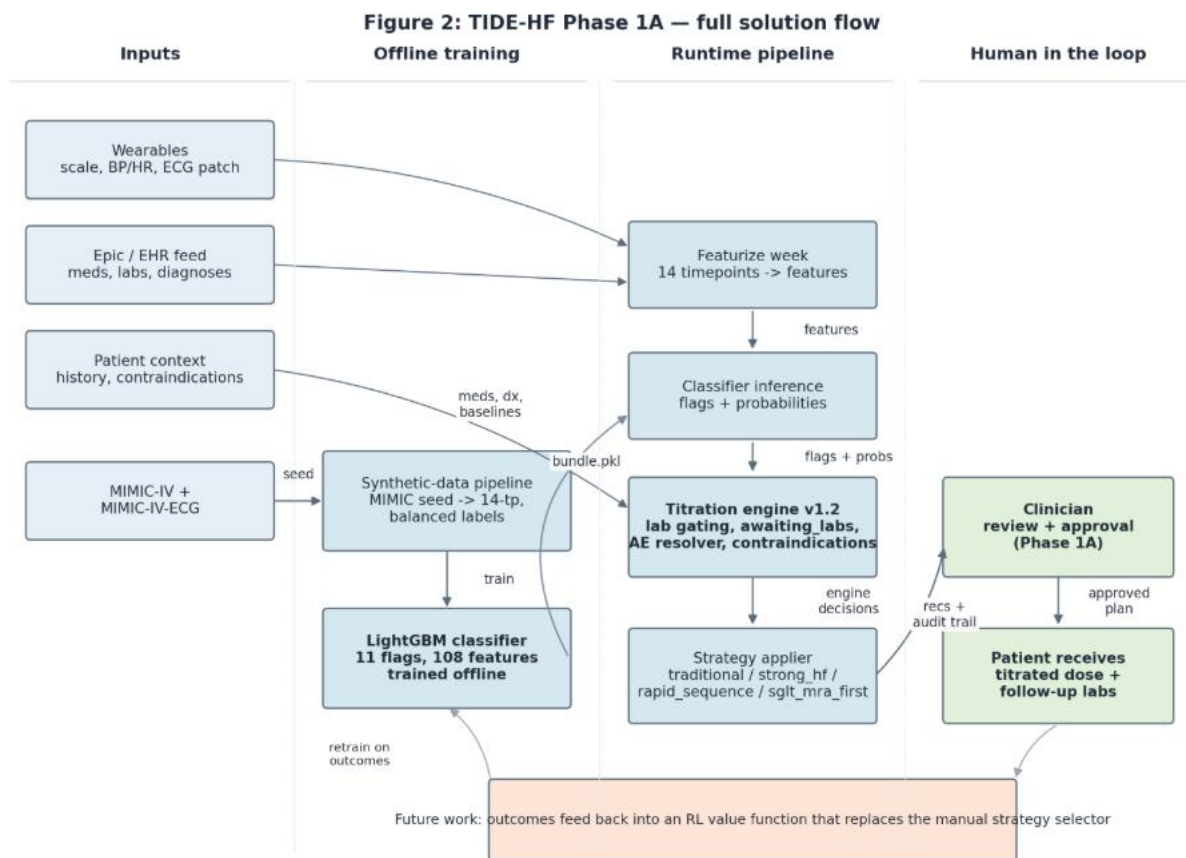
- EHR snapshot panel showing demographics, baseline labs, and current GDMT regimen.
- Active classifier flags and the full eleven-flag probability table.
- Engine summary line: order_labs yes/no, global_stop yes/no, preferred RAAS class, active RAAS class.

- Per-class recommendation table: class, action, dose change, free-text reason.

The UI loads the trained classifier bundle from `models/chf_classifier_lgbm.pkl`. The first launch after install auto-runs the synthetic-data generator and the trainer, so the user only ever runs `python scripts/run_ui.py`.

13. Overall Solution Flow

The diagram below is the full end-to-end picture: the offline training pipeline that produces the classifier bundle, the runtime decision pipeline that consumes wearable + EHR signals, and the clinician-in-the-loop step that gates every medication change in Phase 1A.



Reading the diagram left-to-right: real wearable and EHR signals plus the patient context feed the runtime pipeline (featurize → classifier inference → engine → strategy). The classifier itself was trained offline against the synthetic patient-weeks produced from a MIMIC-IV seed; the trained pickle (about 5–15 MB) is the only artefact carried into runtime. Every recommendation passes through clinician review before reaching the patient. The Streamlit demo connects directly into the runtime pipeline for testing.

The greyed-out arrow is the future-work loop: outcomes data (patient response to titration, any subsequent ER visit, change in NT-proBNP, change in KCCQ score) feeds into a reinforcement-learning value function that eventually replaces the manual strategy selector with a learned policy.

14. MCP Server for Clinical Decision Support

The Model Context Protocol (MCP) server exposes the entire TIDE-HF pipeline — classifier, engine, strategy applier — as a set of structured tools that any MCP-compatible client can call. The motivating use case: a cardiologist or care coordinator already working inside Claude Desktop, Cursor, or a custom MCP client can query patient state and get titration recommendations without opening the Streamlit UI separately.

Architecture

Built on FastMCP from the official mcp Python SDK. The server is a single Python process started with `python scripts/run_mcp.py` (or via the `chf-mcp` console entry). Every tool the server exposes is a simple Python function decorated with `@mcp.tool()`; the decorator handles JSON-schema generation, parameter validation, and the streaming response protocol. Internally each tool calls into the same `evaluate()` / `apply_strategy()` / TideRAG paths that the Streamlit UI and the FastAPI backend use, so the three frontends stay in lockstep.

Server-level instructions enforce two modes

The FastMCP server is initialized with an instructions block that the client exposes to the LLM. It hard-codes the same audience split as the RAG layer:

- Clinician mode (precise medical terminology)
 - Report exact lab values and thresholds ($K > 5.5$ mmol/L, $Cr \Delta > 30\%$, $eGFR$ mL/min/1.73m²).
 - Use drug class names (RAAS, ARNi, MRA, SGLT2i, beta-blocker, loop diuretic).
 - Reference dose ladders and titration ratios (current/target).
 - Cite AE triggers and monitoring schedules.
 - Flag global-stop conditions explicitly with clinical rationale.
- Patient mode (plain language only)
 - Say “heart medicines”, “blood test”, “kidney check”, “water tablet”.
 - Never use: AUROC, titration, RAAS, ARNi, MRA, contraindication, eGFR.
 - Focus on what changes, what to watch for, and when to call the care team.

Tools exposed

Fifteen tools, grouped into clinician-facing decision support and patient-facing explanation. Every tool that takes a preset name validates against the seventeen presets in the synthesizer; invalid names return a clear error listing valid options.

Clinician tool	What it returns
list_patient_presets	All seventeen preset patients with a one-line summary.
get_patient_snapshot	Full clinical snapshot for one preset: demographics, baseline labs, current GDMT, contraindications, eGFR stage.
run_titration	End-to-end pipeline run: featurize → classify → engine → strategy. Returns flags, decisions, dose changes, and any labs ordered.
get_gdmt_optimization_score	Per-class % of target reached and an overall score.
compare_strategies	Run all four strategies on the same preset and report the differences side by side.
check_contraindications	Per-class contraindication check from baseline + diagnoses + meds.
list_strategies	Behavior summary of the four strategies.
get_classifier_info	Bundle metadata, label list, per-flag test-set AUROC.
get_vital_trends	14-timepoint trajectory summary for the preset.
assess_clinical_urgency	Maps engine output to an urgency tier (routine / watchful / urgent / emergency).
get_lab_monitoring_plan	Per-class monitoring schedule (when to recheck K, Cr, etc.).
calculate_risk_score	Composite risk score from active flags + lab values.

Patient tool	What it returns
get_my_medications	Plain-language description of the patient's current GDMT.
get_my_medication_changes	Plain-language explanation of this cycle's changes.

explain_warning_sign

Plain-language description of one classifier flag and what to do.

Embedded clinical knowledge

The server doesn't only proxy the engine — it also bakes in the clinical thresholds and monitoring schedules so the AI client can quote them directly without an extra RAG round-trip. The global-stop thresholds ($K > 6.0$, $Cr \Delta > 30\%$, severe hypotension / bradycardia patterns, composite emergency flag), the per-class lab-AE thresholds (hyperkalemia / AKI / hyponatremia / eGFR cutoffs per drug class), the monitoring schedule (when to recheck labs after a dose change), and the representative drugs per class all live in module-level dictionaries that the tools compose into their replies. This keeps the MCP responses self-contained even when the RAG layer is offline.

Why MCP in addition to the Streamlit UI

Three reasons. First, clinicians who already live inside Claude Desktop or a similar AI assistant can ask follow-up questions that span multiple tools without context-switching to a second app. Second, the structured tool interface makes the system's capabilities discoverable — the MCP client can list all fifteen tools with their parameter schemas, so a user does not have to guess what the system can do. Third, MCP is becoming the default integration surface for the next generation of clinical-AI agents; building this layer now positions TIDE-HF to participate in those workflows without a rewrite.

15. Patient and Clinician RAG Layer

After the engine and strategy applier produce a per-class change record, an optional retrieval-augmented generation (RAG) layer turns those decisions into either a plain-language patient summary or a guideline-cited clinician rationale. Everything runs locally on the laptop — no cloud calls, no API keys, no patient data leaves the machine after the one-time model download.

Stack

Component	Implementation	Notes
Vector store	ChromaDB (persistent, ~50 MB on disk)	tide_hf_v1 collection; cosine similarity
Embeddings	sentence-transformers all-MiniLM-L6-v2 (~80 MB)	CPU-friendly, downloads once on first use
Generation (local)	Ollama serving Mistral 7B (~4.1 GB)	Runs as a Windows service after install

Generation (fallback)	Groq hosted Llama-3.3-70B (cloud)	Activated only when GROQ_API_KEY is set; used if local Ollama unreachable
PDF / DOCX parsing	pypdf + python-docx	Used by the one-time index build

Corpus organization

The RAG corpus lives at the repo root under rag_docs/ and is split into three audience buckets so retrieval can be filtered by who is asking. The current build indexes 606 chunks across these three buckets:

Folder	What's in it	Audience
rag_docs/clinical/	AHA / ACC / HFSA 2022 HF Guidelines, STRONG-HF protocol v3.0, AIM-POWER source paper, drug monographs (carvedilol, empagliflozin, furosemide, spironolactone)	Shown to clinicians
rag_docs/patient/	Beta-blocker patient guides, Entresto patient guide, SGLT2i consumer leaflets, fluid-restriction diet, HF symptom tracker, targeted-symptoms guide	Shown to patients
rag_docs/shared/	Documents that apply to both audiences; queried as a fallback when the audience-specific bucket returns nothing.	Shown to both

Each PDF or DOCX is split into ~400-word chunks with 50-word overlap, embedded via all-MiniLM-L6-v2, and stored in Chroma with audience metadata. The index is idempotent — re-running scripts/setup_rag.py only adds new chunks unless --force is passed.

Two prompts, one engine

The same TideRAG.ask() entry point swaps between two system prompts:

- Patient prompt
 - “Use plain, simple English. Avoid medical jargon. Explain what is happening with the patient's medications and what to watch for. Only use information from the guidelines provided — never invent clinical facts. Keep your answer to 3-4 short paragraphs.”

- Clinician prompt
 - “Give a concise, precise rationale citing AHA 2022 guidelines, STRONG-HF thresholds, and specific lab values where relevant. Use standard medical terminology. Keep your answer under 200 words.”

Retrieval is also audience-aware: a clinical query filters Chroma to clinical + shared chunks; a patient query filters to patient + shared. This stops the patient response from accidentally pulling in dose-titration tables from the AHA guideline that would not make sense in plain language.

Engine-context bridge

`build_engine_context()` converts the TitrationEngine + `apply_strategy` outputs into a plain-text block the LLM consumes alongside the retrieved chunks. The block contains demographics, the global-stop status if active, every active adverse-effect flag, the lab values from this cycle (creatinine annotated with percent change vs. baseline), any labs ordered, the active and preferred RAAS class, and one line per drug class with current dose, new dose, percent of target, and the engine's reason. Awaiting-labs classes from the prior cycle are listed at the bottom. The same builder is used by the Streamlit Ask AI tab and the `/chat` endpoint of the FastAPI backend.

Setup and use

One-time:

- `pip install -e ".[rag]"`
- `ollama pull mistral`
- `python scripts/setup_rag.py` — indexes everything in `rag_docs/` into `rag_db/`

Then:

- `python scripts/run_ui.py` — opens Streamlit, use the “Ask AI” tab
- `python scripts/run_api.py` — exposes `/chat` over HTTP for the website front end
- `python scripts/run_mcp.py` — exposes the same machinery as MCP tools (next section)

If the local Ollama server is unreachable (a known macOS Tahoe Metal issue at the time of writing) and a `GROQ_API_KEY` is set in the environment, the engine transparently falls back to Groq's hosted Llama-3.3-70B. If neither is available, the user sees a single actionable error message describing both fix paths.

16.RAG Model and Map Server Architecture

This project implements a hybrid architecture that combines a **Retrieval-Augmented Generation (RAG) model** with a **Map Server** to deliver accurate, context-aware, and safe clinical decision support for heart failure management. The objective is to generate responses that are not only intelligent and explainable but also grounded in trusted medical knowledge and validated through rule-based safety checks.

The **RAG model** serves as the primary intelligence layer of the system. Unlike traditional language models that rely only on pre-trained knowledge, the RAG model integrates external medical documents with real-time patient context. The knowledge base includes authoritative sources such as the **AHA 2022 Heart Failure Guidelines, STRONG-HF protocol, drug monographs, and patient education materials**, ensuring that responses are evidence-based and clinically reliable.

The RAG pipeline operates in two phases. In the **offline setup phase**, medical documents are loaded, divided into smaller chunks to preserve meaning, and converted into vector embeddings using a sentence-transformer model (all-MiniLM-L6-v2). These embeddings are stored in a vector database (ChromaDB). In the **online query phase**, when a patient or clinician submits a query, the system incorporates patient-specific context such as medications, vitals, lab values, contraindications, and adverse event flags. The query is embedded, and the system retrieves the most relevant document chunks using similarity search. These retrieved documents, along with the patient context and user role, are used to construct a prompt that is passed to a large language model (e.g., Mistral 7B) to generate a final response.

While the RAG model provides flexible and context-aware reasoning, the system also includes a **Map Server**, which acts as a structured, rule-based knowledge layer. The Map Server stores predefined clinical mappings such as drug–side effect relationships, contraindications, lab abnormality risks, and vital sign warning patterns. For example, it can directly identify that hyperkalemia requires caution with MRAs, hypotension affects GDMT titration, or bradycardia may contraindicate beta-blocker use. These mappings allow the system to quickly produce deterministic outputs for well-defined clinical scenarios.

In this project, the **RAG model is used as the primary system**, and the **Map Server is used as a backup mechanism**. The RAG model handles complex queries, guideline-based reasoning, and natural language explanations. However, in situations where retrieval confidence is low, or when critical safety validation is required, the system falls back on the Map Server to ensure correctness and reliability.

This hybrid approach provides the best of both worlds. The RAG model enables **personalized, explainable, and evidence-driven responses**, while the Map Server ensures **consistency, speed, and safety through rule-based validation**. Together, they create a robust and dependable clinical decision-support system suitable for real-world healthcare applications.

17. Future Work

- Reinforcement-learning replacement for the strategy selector. With outcomes data over a long-enough horizon, the four hand-coded strategies become a starting policy that an RL agent improves on. The action space is the same (per-class start / increase / maintain / decrease / stop). The reward is a weighted combination of forward-progress toward target dose,

hospitalization avoidance, and qualitative (possibly KCCQ score) improvement, penalized by adverse events.

- Real wearable integration. The current pipeline emulates wearable input from the synthetic data generator. The next milestone is a thin adapter layer that ingests live data from a scale, BP/HR cuff, and single-lead ECG patch and aggregates it into the same 14-timepoint week.
- Real EHR integration. FHIR pull from the test Epic instance for prescriptions, labevents, and ICD history. Eliminates the manual patient-context entry currently in the UI.
- Clinician-validation study. Run the engine's recommendations past a panel of HF cardiologists on a held-out set of MIMIC visits and measure agreement. The disagreement cases become the next round of guideline refinement.
- Mobile app. The current Streamlit UI is a local demo. The end-state app is mobile-first, with patient-facing alerts ("you should weigh yourself today", "contact your clinician") and clinician-facing review queues.
- Coverage of currently excluded drug classes. Ivabradine and digoxin are present in the AHA guidelines (for more severe cases) but not yet handled by the engine. The contraindication and AE-monitoring logic generalizes; the dose ladder and start-gating need to be added.

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